Lending Club P2P Loan Default Prediction

Team 14 | UE24CS352A - Machine Learning | Problem No. 39
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GitHub: https://github.com/karan21v/LendingClub-P2P-Loan-Default-Prediction

1. Problem Statement

Peer-to-peer (P2P) lenders must estimate repayment risk before funding a loan. Using historical LendingClub data, we build a binary classifier for **Fully Paid vs Default/Charged Off** and a regression model for **Net Annualized Return (NAR)**. The classifier measures credit risk, while predicted NAR supports a simple investment-ranking strategy.

2. Dataset Details

The LendingClub accepted-loan archive is read in chunks and filtered to finalized loans issued during **2012-2015**. Current/late loans are excluded because their final outcome is unknown. Post-loan fields such as total_pymnt and last_pymnt_d are reserved only for constructing historical NAR and are never model inputs.

Property	Observed value
Finalized loans	786,820 (2012: 53,367; 2013: 134,804; 2014: 223,103; 2015: 375,546)
Class balance	640,186 Fully Paid (81.36%); 146,634 Default/Charged Off (18.64%)
Loaded data	33 relevant columns; about 198 MB after filtering
Model inputs	21 numeric + 8 categorical features after engineering
Important features	Grade/sub-grade, interest rate, loan amount, income, DTI, FICO midpoint, credit history, revol_util, account counts

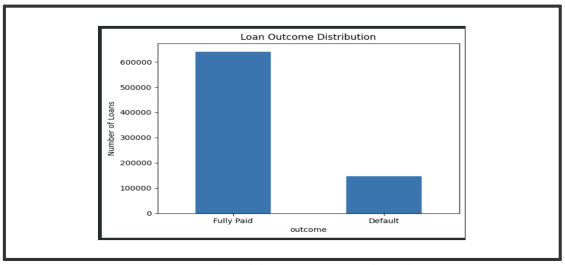


Fig. 1. Finalized-loan outcome distribution.

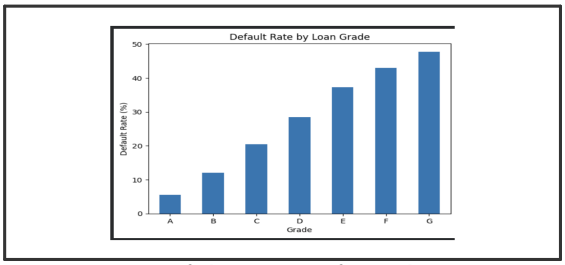


Fig. 2. Default rate increases from grade A to G.

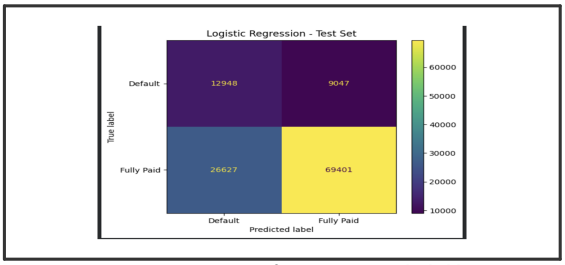


Fig. 3. Logistic Regression confusion matrix on the held-out test set.

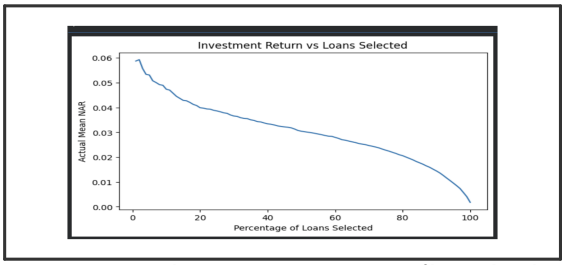


Fig. 4. Historical mean NAR versus percentage of loans selected.

3. Approach

- **Feature engineering.** Term is converted to months; employment length to years; FICO low/high to a midpoint; credit-history age is derived from issue date and earliest credit line; free-text emp_title is dropped.
- **Preprocessing.** Numeric features use median imputation and scaling; categorical features use most-frequent imputation and one-hot encoding inside scikit-learn pipelines.
- **Split and compute.** 70% train / 15% validation / 15% test. Model fitting uses a stratified 200,000-loan sample from the training partition for Colab feasibility.
- **Models.** Dummy, Logistic Regression, MLP and Random Forest are compared for classification; Linear, Ridge, MLP and Random Forest (depth 4/8/10) are compared for NAR regression.

4. Experimental Results and GUI

Classification (test)	Value	Regression / strategy	Observed result
Selected classifier	Logistic Regression	Best validation regressor	Neural Network
Default precision / recall	0.327 / 0.589	Best validation R^2	≈ 0.040
Default F1	0.421	Investment threshold	7.16% predicted NAR
Accuracy / weighted F1	0.698 / 0.726	Ranking result	Top-ranked subsets show higher realized NAR

Interpretation. On the 118,023-loan held-out test set, Logistic Regression correctly detected 12,948 defaults and missed 9,047; 26,627 fully paid loans were falsely flagged. NAR is difficult to predict exactly, but the ranking still carries useful signal because the highest-ranked subsets achieve better realized historical return.

System workflow. TRAIN fits and saves the models once; TEST evaluates the frozen held-out test data and produces the final metrics/graphs; DEMO loads the saved models into the Gradio GUI, where custom borrower values produce live default probability, predicted NAR and INVEST / DO NOT INVEST outputs.

5. Conclusion and Limitations

Accuracy alone is misleading on the 81/19 class distribution; default-class recall and F1 are essential for evaluation. Logistic Regression gives the strongest default-class F1 in our tested setup. NAR is more useful as a ranking signal than as an exact forecast. Main limitations are the historical 2012-2015 cohort, the 200,000-row fitting sample, and use of LendingClub-assigned grade information.

Reference

[1] P. Li and G. Han, "LendingClub Loan Default and Profitability Prediction," Stanford University (supplied reference paper).